E-tiGo Pricing Engine

How It Works & What You Can Control / Updated October 2026

This document walks through how pricing works in the E-tiGo platform, what each setting does, and the business decisions you can make through the admin dashboard without needing any code changes.

How Fares Are Calculated

Every fare in E-tiGo is calculated using a straightforward formula. When a passenger requests a ride, the system looks up the pricing configuration for their city and chosen vehicle class, gets the distance and estimated travel time for the route via Google Maps, and runs this calculation:

> **FARE FORMULA**
>
> **Fare = max(Minimum Fare, Base Fare + (Distance x Per-km Rate) + (Trip Time x Per-minute Rate) + Waiting Charge) x Surge Multiplier − Tier Discount**

The **max()** part is important — it means the fare can never drop below whatever you set as the minimum fare. If a ride is very short (say, 0.5 km down the street), the calculated amount might be lower than the minimum, so the system bumps it up. This protects driver earnings on short trips.

The **Surge Multiplier** is applied after the base fare calculation. When no surge is active, the multiplier is 1.0 (no change). When surge is active (e.g. 1.5x during rush hour), the entire fare is multiplied.

The **Tier Discount** is a loyalty reward: passengers in higher reward tiers get a percentage off the base fare (see Tier Discounts below). Any promo code is taken off after that.

Trip Time is Google's estimated travel time for the route. The fare is **upfront**: the passenger pays the quote, unless they keep the driver waiting or the trip runs significantly longer than quoted (see Upfront Fares below).

What Each Rate Controls

RATE	WHAT IT DOES	BUSINESS IMPACT
Base Fare	Fixed amount charged on every trip, regardless of distance or time.	Sets your floor revenue per ride. Higher base = more predictable earnings per trip.

RATE	WHAT IT DOES	BUSINESS IMPACT
Per-km Rate	Charged per kilometre of the route.	Makes longer trips proportionally more expensive. The primary revenue lever for most rides.
Per-minute Rate	Charged per minute of the quoted travel time (the actual time if the trip runs significantly longer).	Compensates drivers for time in traffic. Higher rates in congested cities make sense.
Minimum Fare	Absolute floor — fare can never go below this.	Protects driver economics on very short trips. Should cover at least fuel + opportunity cost.
Waiting Time Rate	Per-minute charge when the driver is waiting beyond the free period at pickup.	Compensates drivers for idle time. Applied after the free waiting window expires.
Free Waiting Minutes	Grace period before waiting charges begin (default: 5 minutes).	Balances passenger convenience with driver time. Industry standard is 3-5 minutes.

Example Fare Calculation

> **ABUJA — ECONOMY CLASS — 10 KM TRIP, 25 MIN ESTIMATED**

COMPONENT	AMOUNT
Base fare	₦600.00
Distance: 10 km x ₦250/km	₦2,500.00
Time: 25 min x ₦40/min	₦1,000.00
Subtotal	**₦4,100.00**
Surge: 1.0x (no surge)	₦4,100.00
Total fare	**₦4,100.00**

In this example, the calculated fare (₦4,100) is above the minimum fare (₦1,500), so the passenger pays ₦4,100. If this had been a 0.3 km trip with a 2-minute duration, the calculation would give ₦755 — below the minimum — so the fare would be ₦1,500 instead.

With Surge Pricing Active

> **ABUJA — ECONOMY CLASS — 10 KM TRIP, 25 MIN, 1.5x SURGE**

COMPONENT	AMOUNT
Base fare	₦600.00
Distance: 10 km x ₦250/km	₦2,500.00
Time: 25 min x ₦40/min	₦1,000.00
Subtotal	**₦4,100.00**
Surge: 1.5x	₦6,150.00
Total fare	**₦6,150.00**

Recommended Default Rates

These are the recommended starting rates based on current fuel costs and market analysis (September 2026):

Abuja

SETTING	ECONOMY	COMFORT	PREMIUM
Base Fare	₦600	₦800	₦1,200
Per-km Rate	₦250	₦350	₦500
Per-minute Rate	₦40	₦55	₦80
Minimum Fare	₦1,500	₦2,000	₦3,000
Waiting Rate	₦50/min	₦65/min	₦100/min
Free Waiting	5 min	5 min	5 min

Lagos

SETTING	ECONOMY	COMFORT	PREMIUM
Base Fare	₦600	₦800	₦1,200

SETTING	ECONOMY	COMFORT	PREMIUM
Per-km Rate	₦250	₦350	₦500
Per-minute Rate	₦40	₦55	₦80
Minimum Fare	₦1,500	₦2,000	₦3,000
Waiting Rate	₦50/min	₦65/min	₦100/min
Free Waiting	5 min	5 min	5 min

These rates ensure drivers profit on every trip at current fuel costs (₦1,430/litre). Example: a 10 km, 25 min Abuja Economy trip yields ₦4,100 — with fuel cost around ₦1,590 for a sedan, that leaves ₦2,510 for the driver before platform commission.

You can adjust these at any time through the admin dashboard. They are defaults — set them as starting points and tune based on your market data.

Waiting Time Charges

When a driver arrives at the pickup location and the passenger isn't ready, waiting time charges kick in — but only after a free grace period. Here's how it works:

> **WAITING CHARGE FORMULA**
>
> **Waiting Charge = max(0, Actual Wait - Free Waiting Minutes) x Waiting Time Rate**

Every pricing config includes a **free_waiting_minutes** value (defaults to 5 minutes). The driver waits without the passenger being charged anything for that period. After the free period expires, every additional minute is charged at the **waiting_time_rate**.

Example: Waiting Time on an Abuja Economy Ride

> **DRIVER WAITS 8 MINUTES — 5 MIN FREE, RATE: ₦50/MIN**

Total wait time	8 min
Free waiting period	-5 min
Chargeable minutes	3 min
Waiting charge: 3 min x ₦50/min	**₦150.00**

> **How estimates work:** When passengers request a fare estimate, they see the waiting time policy (free minutes and per-minute rate) so they know the rules before booking. The actual waiting charge is only applied when the ride is completed, based on how long the driver actually waited.

Upfront Fares: Quote vs. Final Fare

E-tiGo uses **upfront fares**, the same approach as Uber and Bolt: the price the passenger sees before booking is the price they pay. Google's travel-time estimate already allows for normal traffic, so ordinary delays don't change the fare. Only two things can change it:

1. **Waiting:** if the driver waits beyond the free minutes at pickup, the waiting charge is added.
2. **A much longer trip:** if the trip runs **more than 25% longer than quoted, and at least 10 minutes longer**, the time charge is worked out again using the actual trip time (from the PIN start to the end of the trip).

If the trip is quicker than expected, the passenger still pays the quote.

LOCKED IN AT BOOKING	CAN CHANGE DURING THE TRIP
Pricing rates (the version in effect when the ride was booked)	**Waiting time:** from the driver's arrival to trip start
Distance (Google's route from pickup to destination)	**Trip time:** only when the trip runs significantly longer
Surge multiplier	
Tier discount	
Promo code (if valid at booking)	

So a surge that ends mid-trip, or a price change made while the passenger is riding, never changes their fare.

Protection Against Trips Left Running

If a driver forgets to end a trip, the clock would keep running. To protect passengers, the billed trip time is **capped at 3× the quoted time**. Any ride that hits the cap is flagged for the operations team to review.

Examples

> **ABUJA — ECONOMY CLASS — 10 KM TRIP, QUOTED AT 25 MIN (₦4,100)**

For a 25-minute quote, the tolerance is 10 minutes (25% of 25 minutes is only 6¼, so the 10-minute minimum applies), and the cap is 75 minutes.

TRIP ACTUALLY TOOK	WHAT HAPPENS	TIME BILLED	FINAL FARE
18 min	Quicker than quoted: the quote stands	25 min	**₦4,100**
32 min	Within the 10-minute tolerance: the quote stands	25 min	**₦4,100**
40 min	15 minutes over: actual time billed	40 min	**₦4,700**
3 hours (trip not ended)	Over the cap: billed at the cap and flagged for review	75 min	**₦6,100**

Fare = ₦600 base + 10 km × ₦250 + minutes billed × ₦40.

These three settings (25%, 10 minutes, 3×) are server settings and can be changed without an app release.

Surge Pricing

Surge pricing lets you increase fares during periods of high demand. The system applies a multiplier (e.g. 1.5x) to the calculated fare. You control when surge is active, how high it goes, and what triggers it.

How It Works

The surge multiplier is applied **after** the base fare calculation (including the minimum fare check):

> **Surged Fare = Fare (after minimum check) x Surge Multiplier**

When passengers see a fare estimate during an active surge, they see:

- The total fare (already multiplied)
- A "1.5x surge" indicator so they know pricing is elevated
- The surge rule name (e.g. "Morning Rush Hour")

Three Types of Surge Rules

TYPE	HOW IT WORKS	BEST FOR
Manual	You toggle it on/off from the admin dashboard. Active immediately when enabled.	Rain, accidents, special events, emergencies — anything you can see happening and want to respond to quickly.
Time-Based	Fires automatically based on a schedule: which days of the week, start time, and end time.	Rush hours (e.g. Mon-Fri 7-9am and 5-8pm), weekend night surges (Fri-Sat 10pm-2am).
Demand-Based	Fires automatically when passengers asking for rides outnumber free drivers by the ratio you set.	Real-time demand spikes, e.g. a concert ending or a sudden downpour.

What You Control

For each surge rule, you set:

SETTING	WHAT IT DOES	CONSTRAINTS
City	Which city the rule applies to.	Required.
Vehicle Class	Optionally scope to a specific vehicle class. Leave empty = applies to all classes in the city.	Optional.
Name	A label you'll see in the dashboard (e.g. "Morning Rush Hour", "Heavy Rain").	Required.
Multiplier	How much to multiply the fare by. 1.5 = 50% increase, 2.0 = double.	1.00 to 5.00.
Priority	When multiple rules match at the same time, the highest priority wins.	0-100 (higher = wins).
Effective From / Until	Date range when the rule can be active.	"Until" is optional — omit for no expiry.
Active	Master on/off switch.	Toggle anytime.

Time-Based Schedule

For time-based rules, you configure:

- **Days of week:** 1 = Monday, 7 = Sunday. Example: `[1, 2, 3, 4, 5]` for weekdays.
- **Start time:** When surge begins (24-hour format, e.g. `07:00`).
- **End time:** When surge ends (e.g. `09:00`).

Overnight spans work too — setting start `22:00` and end `06:00` covers the late-night period.

Demand-Based Settings

For demand-based rules, the system counts the passengers who requested a ride in the last 10 minutes and the drivers who are online and free right now in that city (and vehicle class, if the rule has one). You configure:

- **Demand-to-supply ratio:** how many passengers per free driver before surge starts. Example: `2.0` means surge when twice as many passengers are asking as there are free drivers.
- **Minimum requests (optional):** the fewest passengers who must be asking before the rule can fire (default 5). This stops a quiet moment, say 2 requests and 1 free driver, from triggering surge.

Demand is re-checked every minute, so surge switches off on its own once drivers catch up.

Priority & Conflict Resolution

If a city has multiple active surge rules that all match at the same time (say, a rush hour rule AND a manual rain rule), the system picks the one with the **highest priority number**. If two rules have the same priority, it picks the one with the **higher multiplier**.

This means you can layer rules: set a weekday rush hour at priority 5 (1.3x) and a manual "heavy rain" at priority 10 (1.8x). On a rainy Monday at 8am, both match — the rain rule wins because it has higher priority.

Example Surge Scenarios

Abuja Morning Rush Hour:

- Type: Time-based
- Days: Monday-Friday
- Time: 07:00-09:00
- Multiplier: 1.3x
- Priority: 5

Heavy Rain (Manual):

- Type: Manual
- Multiplier: 1.8x
- Priority: 10
- Toggle on when it starts raining, toggle off when it stops.

Lagos Demand Spike:

- Type: Demand-based
- Ratio: 2.0 (twice as many passengers asking as free drivers)
- Minimum requests: 8
- Multiplier: 1.4x
- Priority: 7

Weekend Night Premium:

- Type: Time-based
- Days: Friday, Saturday
- Time: 22:00-03:00
- Multiplier: 1.5x
- Priority: 5
- Vehicle class: Premium only

What Passengers See

When surge is active, the fare estimate response includes:

```
"surge": {
  "active": true,
  "multiplier": 1.5,
  "rule_name": "Morning Rush Hour"
}
```

The mobile app should display a clear surge indicator so passengers can decide whether to wait or accept the higher fare. When no surge is active, the multiplier is 1.0 and the fare is unchanged.

Tier Discounts

Passengers earn reward points on every ride and move up through reward tiers. Each tier takes a percentage off the **base fare** of every ride:

TIER	DISCOUNT ON BASE FARE	ON A ₦600 ECONOMY BASE FARE
Bronze	0%	₦0
Silver	2%	₦12
Gold	5%	₦30
Platinum	10%	₦60
Diamond	15%	₦90

The discount is applied after surge, is locked in when the passenger books, and is never more than the fare itself. The quote the passenger sees already includes it. You can change the percentages in the rewards settings of the admin dashboard.

> **Example:** A Gold passenger takes the 10 km, 25 min Abuja Economy trip with no surge: ₦4,100 − ₦30 (5% of ₦600) = **₦4,070**.

Pricing by City and Vehicle Class

Pricing is configured per **city + vehicle class** combination. This means you can set completely different rates for:

- Different cities (Lagos pricing vs. Abuja pricing vs. Port Harcourt pricing)
- Different vehicle classes within the same city (Economy vs. Comfort vs. Premium in Lagos)

This gives you fine-grained control. You might want Economy rides in Abuja to start at ₦600, but Premium rides to start at ₦1,200 — and those same classes in Lagos might have completely different numbers based on local economics.

> **Key point:** You need to create a pricing config for every city + vehicle class combination you want to offer rides in. If a city has 3 active vehicle classes, that's 3 pricing configs to set up. Without a pricing config, that vehicle class won't appear in fare estimates for that city.

Versioning — How Pricing Changes Work

Pricing configs are **versioned and append-only**. When you create a new pricing config for the same city + vehicle class, the system doesn't overwrite the old one — it creates a new version alongside it. This is deliberate, and it gives you two important capabilities:

Price History

You can always look back and see what your rates were at any point in time. This is useful for auditing, resolving disputes, and understanding revenue trends. The admin dashboard shows you all versions with their effective dates.

Scheduled Price Changes

Every pricing config has an **effective_from** date. You can create a new pricing config today that won't kick in until next Monday. The system will keep using the current rates until that date, then automatically switch to the new ones. No manual intervention needed on the day.

> **In-progress ride protection:** When a passenger books a ride, the system takes a snapshot of the pricing at that moment and locks it to that ride. If you change pricing while someone is mid-trip, their fare stays on the rate they were quoted. This prevents disputes and builds passenger trust.

Cross-City Rides & Edge Cases

The pricing engine handles several real-world scenarios automatically. Here's how they work and what the passenger sees.

Cross-City Rides

When a passenger books a ride from one city (e.g. Abuja) to another city (e.g. Lagos), the system:

1. **Detects the destination city** automatically using the destination coordinates
2. **Applies pickup city pricing** — the ride uses Abuja rates, not Lagos rates

3. **Warns the passenger** with a message like: *"This is a cross-city ride from Abuja to Lagos. Pickup city (Abuja) pricing applies for this trip."*

This is the industry standard approach (used by Uber, Bolt, and others). The pickup city's rates apply for the entire ride because that's where the driver started and where the platform's operational costs are based.

> **Example:** A passenger in Abuja books a ride to Lagos (850 km). The Economy fare would be: ₦600 base + (850 × ₦250/km) + (600 min × ₦40/min) = ₦237,100 — using Abuja rates.

Destination Outside Service Area

If the destination doesn't fall within any of E-tiGo's active cities, the estimate still works — it uses the pickup city's pricing. The app shows a warning: *"Your destination is outside our current service areas. The driver may not be able to accept return trips from the destination."*

This lets passengers take rides to locations you haven't launched in yet, while managing expectations about driver availability for the return trip.

Long-Distance Rides

Routes over 100 km automatically include a warning that the fare is estimated and may vary. The fare is still calculated normally — this is just a heads-up for the passenger that real-world conditions (traffic, road closures, detours) may affect the final fare more on longer routes.

Short Rides & Minimum Fare

For very short rides (e.g. 0.3 km), the calculated fare might be much lower than the minimum fare. The system automatically bumps it to the minimum. This protects driver earnings — a trip that calculates to ₦755 would be charged at ₦1,500 (the minimum) instead.

What the App Sees

When edge cases are detected, the API response includes extra fields:

```
"warnings": [
  "This is a cross-city ride from Abuja to Lagos. Pickup city (Abuja) pricing applies for this trip.",
  "This is a long-distance ride (over 100 km). Fare is estimated based on route distance and may vary."
],
"cross_city": {
  "destination_city_id": 2,
  "destination_city_name": "Lagos",
  "pickup_city_name": "Abuja"
}
```

The mobile app should display these warnings clearly before the passenger confirms the booking. For standard same-city rides, these fields are not included.

Business Decisions You Can Make

Here are the main levers you have through the admin dashboard, and when you might want to pull each one:

Adjust Rates by City

If a particular city has higher fuel costs, more traffic, or a different cost of living, you can set rates accordingly. You might charge more per km in a city with expensive fuel and more per minute in a city with heavy congestion.

Differentiate Vehicle Classes

Premium and luxury classes should carry a meaningful price premium. A rule of thumb: Comfort is typically 1.3-1.5x Economy, Premium is 2-2.5x. EV classes could be priced at a slight discount to incentivise green rides, or at a premium if positioned as a luxury offering.

Schedule Rate Changes

Planning a rate increase? Create the new pricing config with a future effective date. Want to run a promotional week with lower fares? Set lower rates effective Monday and higher rates effective the following Monday. Both will apply automatically.

Set Minimum Fares Strategically

The minimum fare is your most important tool for driver retention. If it's too low, drivers lose money on short trips and stop accepting them. If it's too high, passengers walk instead of booking. Check your average trip distance — minimum fare should be attractive enough that drivers don't skip short rides.

Configure Waiting Time Policy

You control both the free waiting period and the per-minute rate. The default is 5 minutes free (Bolt uses 4). Setting this too low frustrates passengers; too high and drivers lose money waiting around. You can adjust per city and vehicle class.

Use Surge Pricing

Activate surge during periods of high demand to balance supply and demand. A moderate surge (1.2-1.5x) encourages more drivers to come online while slightly reducing ride requests — bringing the market toward equilibrium. Use time-based rules for predictable peaks, manual rules for unexpected events, and demand-based rules to react automatically when requests outpace drivers.

Launch a New City

When expanding to a new city: create the city in the admin panel, enable the vehicle classes you want to offer there, then create pricing configs for each city + vehicle class combination. The city goes live for passengers as soon as pricing is in place.

Feature Status

Here's where each part of the pricing engine stands:

FEATURE	WHAT IT DOES	STATUS
Surge Pricing	Dynamic multiplier during high demand. Three rule types: manual, time-based, demand-based.	**Done**
Google Maps Integration	Fare estimates use actual road distances and real-time traffic data via Google Maps Distance Matrix API.	**Done**
Cross-City Ride Detection	Automatic detection of cross-city rides with warnings and pickup city pricing. Handles destination outside service area and long-distance rides.	**Done**
Demand-Based Surge	Demand-based surge rules fire automatically from live demand against free drivers.	**Done**
Upfront Fares	The passenger pays the quote. The time charge is recalculated only when a trip runs significantly longer, with a cap that protects against trips left running.	**Done**
Promo Integration	Promo codes are checked at booking and taken off the final fare at the end of the trip.	**Done**
Tier Discounts	Passengers in higher reward tiers get a percentage off the base fare. Configured through the rewards settings.	**Done**

Quick Reference

Admin Actions — Pricing

ACTION	WHERE IN DASHBOARD
Create new pricing	Admin → Pricing → New Config
View all pricing history	Admin → Pricing (filter by city / vehicle class)
Check current active rate	Admin → Pricing → Current (select city + class)
Schedule a future rate	Create new pricing with a future "Effective From" date

Admin Actions — Surge Pricing

ACTION	WHERE IN DASHBOARD
Create surge rule	Admin → Surge Rules → New Rule
View all surge rules	Admin → Surge Rules (filter by city)
Toggle surge on/off	Admin → Surge Rules → Toggle Status
Check current multiplier	Admin → Surge Rules → Current Multiplier (select city)
Set rush hour schedule	Create time-based rule with days + times
Emergency surge	Create manual rule, toggle on immediately

E-tiGo Backend — Pricing & Surge Engine Documentation Questions? Reach out and we'll walk you through any of this in more detail.